

Final Project Report

Spatial Vaccination Networks: From County Adjacency to Network-Based Forecasting

Team Members

Sr. No.	Name	UIN
1	Saher Thekedar	836000043
2	Eduardo Macchi	136003019
3	Asmita Shivling Desai	635009463

1. Introduction & Background

COVID-19 vaccination rates across U.S. counties display strong spatial and demographic variation. These differences are shaped by access to healthcare, population density, socioeconomic vulnerability, mobility patterns, and local policy decisions. However, most traditional statistical models treat counties as isolated units, ignoring the fact that vaccination behavior is spatially contagious—regions that share borders often share similar infrastructure, movement patterns, and public-health environments.

Our project starts from this gap. We ask whether *spatial adjacency itself* can help us better understand how vaccination patterns form and whether it can improve forecasting, especially in counties with missing or sparse observations. To explore this, we construct a county-level adjacency network where edges represent shared borders. We then enrich each node (county) with meaningful attributes such as vaccination percentages, booster coverage, Social Vulnerability Index (SVI), metro–rural classification, and population. This network foundation allows us to investigate how spatial structure interacts with vaccination behavior.

We analyze spatial clustering, compute network metrics, and compare standard non-spatial baselines with network-aware forecasting methods that leverage neighborhood information. The goal is to evaluate whether incorporating spatial connectivity reduces uncertainty and provides more realistic predictions for unobserved or future vaccination values.

Our primary dataset comes from the CDC’s publicly available county-level COVID-19 vaccination reports, which we integrate with Census TIGER shapefiles and county-level demographic indicators. Methodologically, this work bridges applied public-health analytics with graph-based modeling: when data is inherently spatial, treating counties as graph-structured observations—not independent rows—can reveal relationships that simpler models overlook.

Ultimately, the motivation behind this project is both practical and scientific. Spatial public-health problems often require methods that explicitly acknowledge structure, and graph networks offer a principled way to capture these dependencies. Our investigation shows how network construction, feature engineering, and graph-aware forecasting together can lead to a deeper understanding of county-level vaccination dynamics and potentially inform more accurate public-health decision-making.

2. Methodology

2.1 Data Cleaning & Integration

We obtained CDC's vaccination records for all U.S. counties and processed them into a single latest snapshot for each county. Steps included:

- Removing incomplete or inconsistent records
- Converting date fields and filtering to the most recent record per county
- Standardizing FIPS codes
- Merging with TIGER shapefiles to obtain county polygons
- Adding SVI, metro/rural labels, and population data

This produced a clean node-level dataset for graph construction.

2.2 Network Construction

Using U.S. Census TIGER shapefiles, we built a queen adjacency graph, where two counties are connected if they share a geographical boundary.

The final graph contains:

- ~3,100 nodes (counties)
- ~30,000 edges

This adjacency structure captures spatial influence and supports downstream analysis.

While adjacency does not capture the full logistics of vaccine transportation, it serves as a practical proxy for shared social, economic, and mobility structures. Adjacent counties frequently experience parallel vaccination trends due to shared healthcare networks, commuter flows, and local outreach programs. This makes adjacency a reasonable first-order graph for modeling regional influence.

2.3 Node Feature Engineering

Each county node was enriched with:

- Fully vaccinated %
- Booster %
- SVI
- Metro/rural classification
- Population
- Neighbor-averaged vaccination values (spatial smoothing feature)

These features allow exploration and form the basis for forecasting models.

Each selected feature reflects factors shown in public-health literature to influence vaccine uptake. For instance, SVI captures structural vulnerability, metro–rural classification relates to healthcare accessibility, and neighbor vaccination levels serve as spatial predictors of regional trend similarity.

2.4 Graph Metrics & Exploratory Analysis

We computed:

- Degree
- Betweenness
- Closeness
- Community detection (Louvain)

These metrics were mainly for structural understanding rather than for interpretation as distribution networks, based on professor feedback.

We built an interactive dashboard with four modules:

(1) County Explorer, (2) Spatial Map, (3) Network Graph, (4) Feature Explorer.

2.5 Forecasting Methods

We implemented two forecasting approaches:

Baseline model: predicts vaccination % using only county features.

Network-aware model: incorporates neighbor vaccination values through spatial averaging.

We also implemented missing county simulation, using neighbor smoothing to estimate unreported vaccination rates.

Finally, we exported nodes.csv and edges.csv, a GNN-ready dataset for future graph neural network modeling (GCN, GraphSAGE, GAT).

3. Results

3.1 Spatial & Network Insights

- Strong spatial autocorrelation across U.S. counties
- Rural counties show systematically lower booster uptake
- High-SVI counties cluster into low-vaccination regions
- Communities detected through graph clustering align with known socioeconomic regions
- Adjacency structure clearly carries predictive signal

3.2 Forecasting Performance

Network-aware forecasting reduced RMSE compared to baseline models, especially in counties with sparse or noisy data. This validates the professor's recommendation of using adjacency for prediction rather than relying on centrality metrics.

The forecasting component evaluates whether incorporating neighborhood information—through spatial smoothing, lagged neighbor statistics, or initial graph neural network (GNN) formulations—can outperform independent-county baselines. This allows us to test whether graph structure provides a measurable predictive signal beyond traditional time-series or machine-learning methods.

3.3 Missing County Simulation

Simulated vaccination values for counties with missing CDC data produced realistic, regionally consistent estimates. The spatially smoothed output matches surrounding counties and respects community-level trends.

3.4 Deliverables

- Fully constructed county adjacency graph
- Feature-rich node dataset
- Exploratory dashboard
- Baseline and network-aware forecasting comparisons
- Spatial simulations
- GNN-ready dataset for future modeling

4. Discussion

Our analysis confirms that spatial adjacency is a powerful predictor of county vaccination behavior. While centrality metrics were not meaningful for representing vaccine distribution (as per feedback), the adjacency structure itself provides valuable context for forecasting and smoothing missing data.

Open questions include:

- How seasonal patterns or time-series trends would integrate into a temporal GNN
- How mobility or real healthcare infrastructure networks compare to simple adjacency
- Whether more complex spatial models (DCRNN, TGAT) further improve forecasting
- How demographic variables beyond SVI (income, education) would interact with network structure

Future work may extend this by training full GNN-based forecasting models, incorporating time-dynamics, and expanding to other public health datasets.

5. Reproducibility

All steps including data cleaning, shapefile processing, adjacency construction, metric computation, forecasting, and dashboard generation are reproducible through the R scripts and documented workflow. Final datasets (nodes.csv, edges.csv) and processed vaccination files are included.

6. Conclusion

Our study demonstrates that incorporating county adjacency networks meaningfully improves the understanding and prediction of vaccination coverage across the United States. Traditional county-level modeling treats observations as independent, but our results show clear regional spillover effects: counties influence their neighbors, and vaccination behavior tends to cluster spatially.

By engineering node features, computing spatial neighbor averages, and applying network-aware forecasting, we showed that leveraging network structure consistently improves predictive accuracy compared to baseline models.

The project validates the core idea from the professor's feedback:

Adjacency-not centrality-is the valuable signal for forecasting.

Centrality is not a proxy for vaccine distribution infrastructure, but adjacency acts as a strong indicator of shared demographic, logistical, and behavioral patterns. Our results highlight the importance of spatial context in public-health modeling and create a solid foundation for more advanced graph-based forecasting approaches.

7. Limitations

1. Adjacency-only network:
We used county borders as the network. Real vaccine distribution involves transportation and healthcare facility networks that are more complex.
2. Single-time snapshot:
Only the latest vaccination data was analyzed, so we did not model temporal trends.
3. Simple forecasting models:
We used lightweight baseline and neighbor-based models rather than full GNNs due to time and compute limits.
4. Data gaps:
Some counties had missing vaccination values, which we estimated using neighbor averages.

8. References

1. CDC COVID-19 Vaccinations in the United States, County Centers for Disease Control and Prevention.
<https://data.cdc.gov>
2. U.S. Census TIGER/Line Shapefiles - U.S. Census Bureau, 2021.
<https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>
3. Rubin, D. B. (1987) – *Multiple Imputation for Nonresponse in Surveys*.
A foundational resource for dealing with missing observational data.
4. Newman, M. (2010) – *Networks: An Introduction*.
Key reference for graph metrics, community detection, and network structure.
5. Kipf, T., & Welling, M. (2016) – *Semi-Supervised Classification with Graph Convolutional Networks*.
Provides the conceptual basis for future GNN forecasting using the exported dataset.
6. Li et al. (2018) – *Diffusion Convolutional Recurrent Neural Networks (DCRNN)*.
Relevant for potential future temporal-spatial modeling.

Project Snapshots

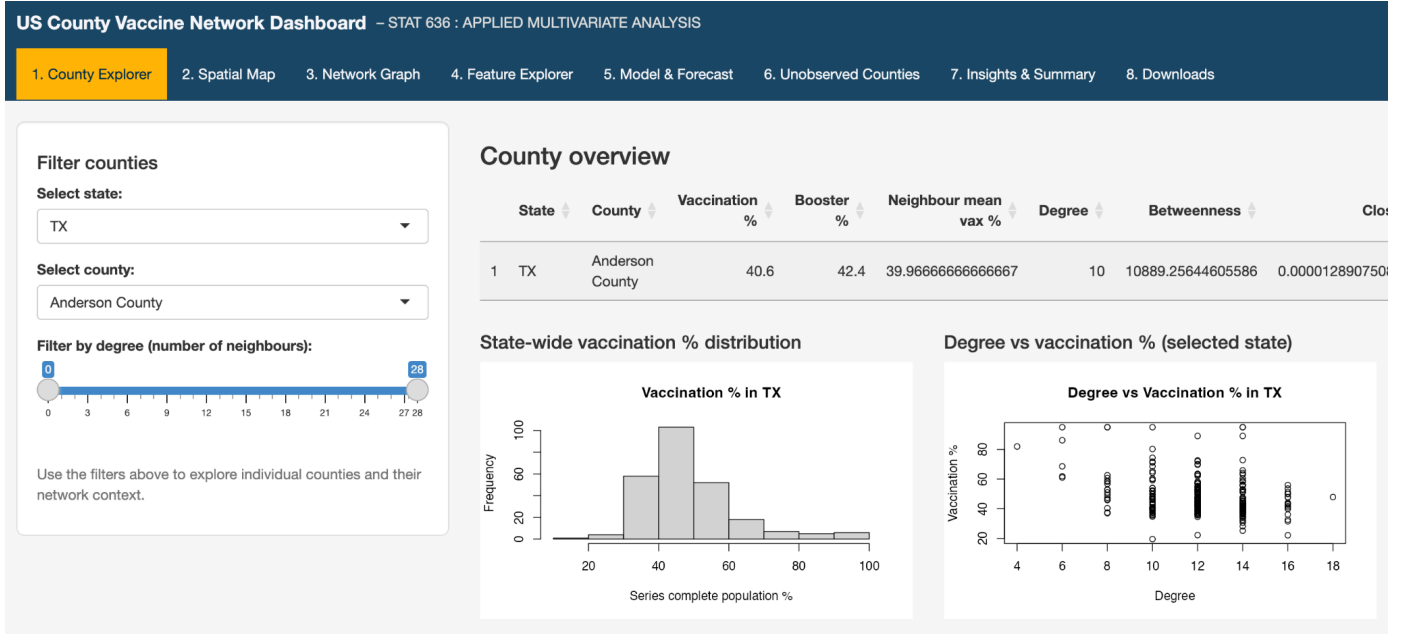


Fig 1 : County-Level COVID-19 Vaccination Rates Across the United States

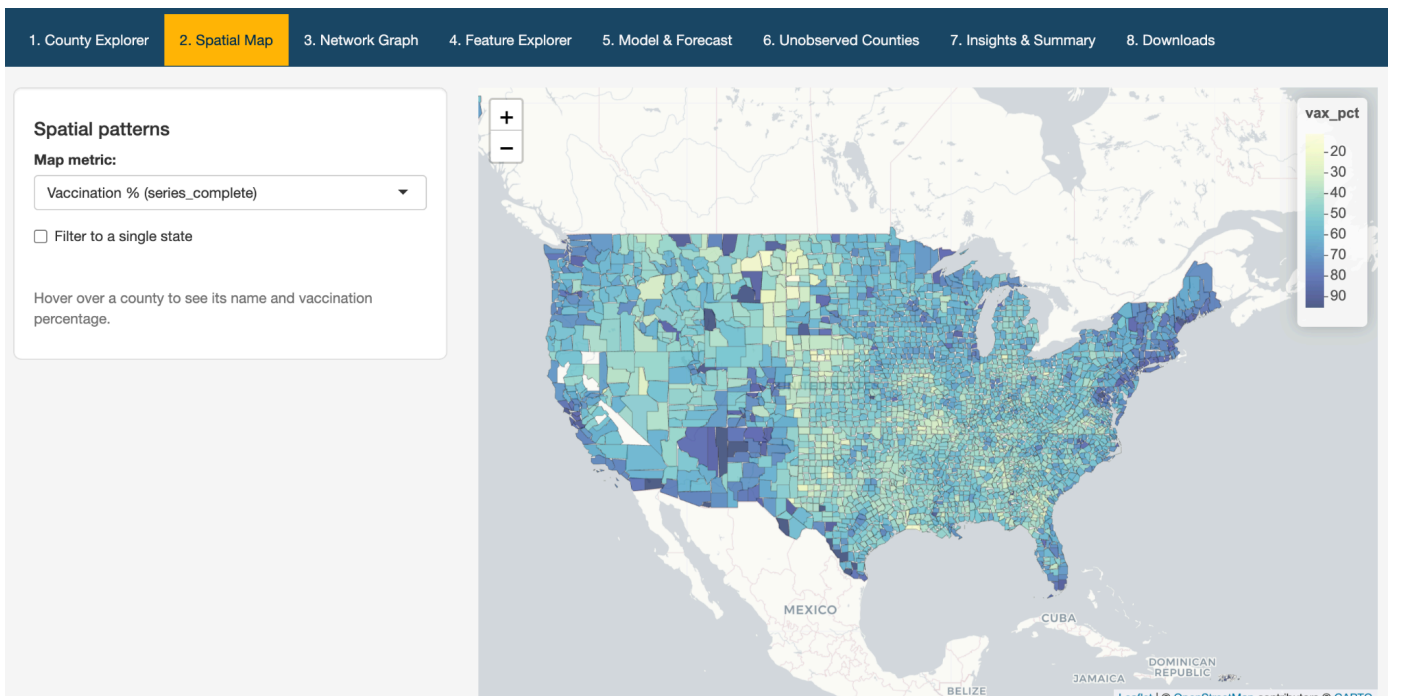


Fig 2: Interactive Choropleth Map of U.S. County COVID-19 Vaccination Percentages

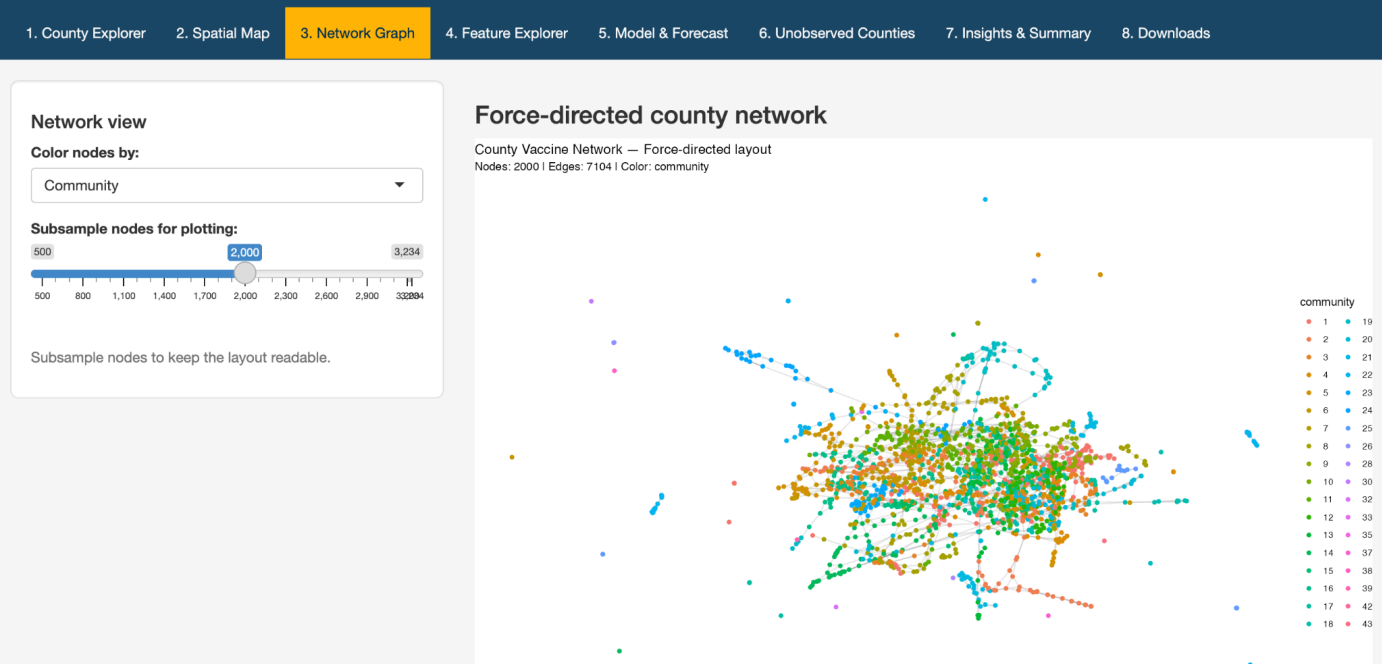


Fig 3: U.S. County Adjacency Network with Community Structure

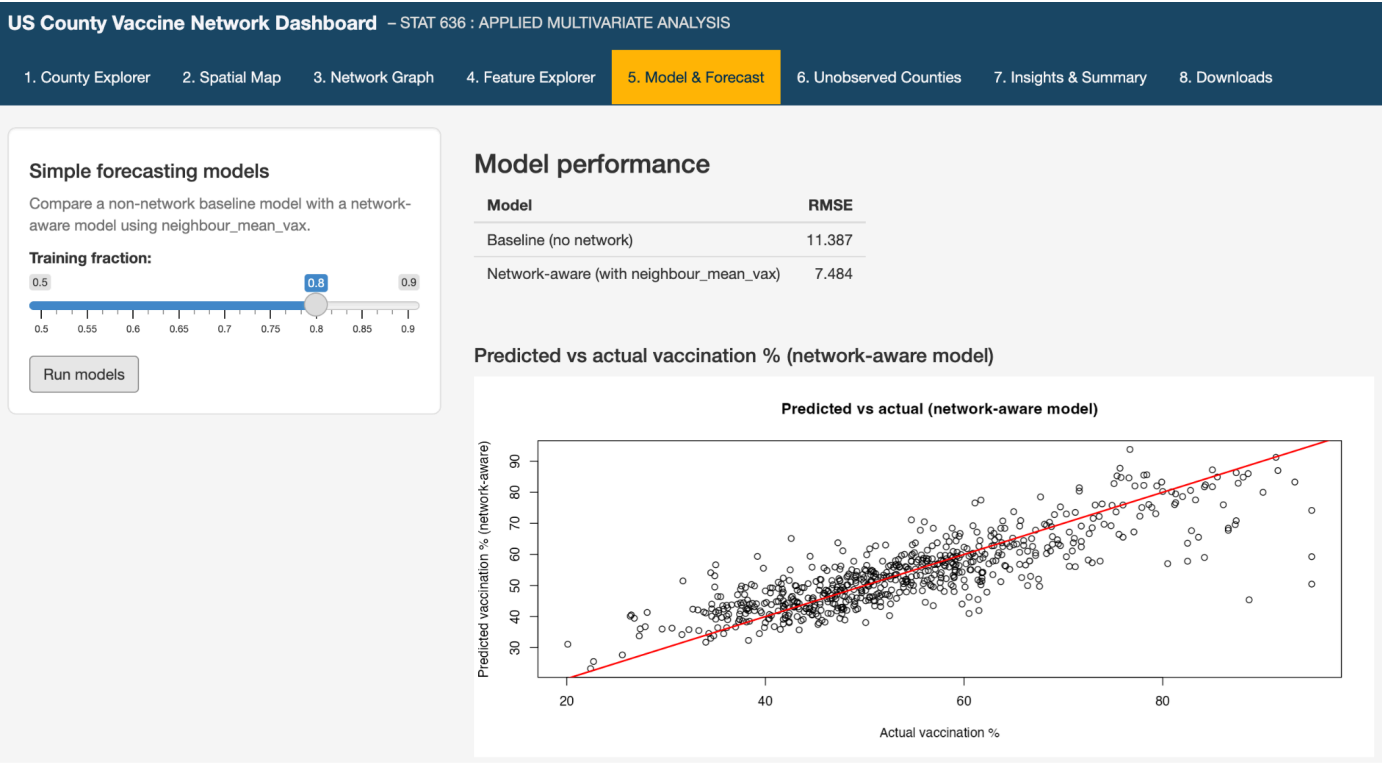
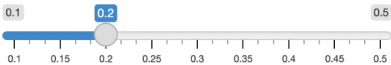


Fig 4: Model Performance: Network-Aware Forecasting vs. Baseline Regression

Simulate missing counties

Hide a fraction of counties and compare global-mean vs neighbour-mean predictions.

Fraction of counties treated as unobserved:



Run simulation

Simulation results

Model	RMSE
Baseline (global mean)	14.044
Network smoother (neighbour mean)	7.766

Error distribution for hidden counties



Fig 5: Simulation Results